

The Bill Belongs to the Fan, Not the Path

Let

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}\right),$$

so that

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}, \quad \mathbb{E}P = \mathbb{E}Q = 2.$$

An exact depth- n zero-repair tree is a family $\{B_h\}_{|h| \leq n}$ satisfying

$$Q * B_h = \sum_x P(x) \delta_x * B_{hx}. \quad (1)$$

Let $\mathcal{K}_r$ be the set of feasible roots at remaining depth r and

$$\mathcal{B}_r := \inf_{B \in \mathcal{K}_r} \mathbb{E}B, \quad \mathcal{B}_0 := 0.$$

If h has $r = n - |h|$ steps remaining, define its *frontier defect*

$$\Delta_h := \mathbb{E}B_h - \mathcal{B}_r \geq 0.$$

Theorem 1 (Forced counterfactual bill). *For every exact tree, every node h with r steps remaining, and every $0 \leq s \leq r$,*

$$\boxed{\sum_{|g|=s} P(g) \Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.} \quad (2)$$

Moreover, for the quartic pair above, optimality itself forces

$$\boxed{n = 3 : \quad \Delta_{(4)} = 0,} \quad (3)$$

for every optimal depth-3 tree, and

$$\boxed{n = 4 : \quad \Delta_{(4)} = \Delta_{(4,4)} = 0,} \quad (4)$$

for every optimal depth-4 tree.

Consequently every depth-3 optimum satisfies

$$\boxed{\mathcal{B}_3 - \mathcal{B}_2 = \sum_{x \neq 4} P(x) \Delta_x,} \quad (5)$$

and every depth-4 optimum satisfies

$$\boxed{\mathcal{B}_4 - \mathcal{B}_3 = \sum_{x \neq 4} P(x) \Delta_x, \quad \mathcal{B}_3 - \mathcal{B}_2 = \sum_{x \neq 4} P(x) \Delta_{(4,x)}.} \quad (6)$$

Thus the forced contact path carries zero excess; the entire marginal bill is carried by its counterfactual siblings.

Proof. Taking expectations in (1) and using $\mathbb{E}P = \mathbb{E}Q$ gives

$$\mathbb{E}B_h = \sum_x P(x) \mathbb{E}B_{hx}.$$

Iterating s generations,

$$\mathbb{E}B_h = \sum_{|g|=s} P(g) \mathbb{E}B_{hg}.$$

Since

$$\mathbb{E}B_h = \mathcal{B}_r + \Delta_h, \quad \mathbb{E}B_{hg} = \mathcal{B}_{r-s} + \Delta_{hg},$$

and $\sum_{|g|=s} P(g) = 1$, identity (2) follows immediately.

It remains only to show that the contacts in (3)–(4) are forced on the *entire* optimal face, rather than selected by one LP vertex.

For $n \leq 4$ the problem is a finite rational linear program. Indeed, conditional on any root atom $J_0 > 4n$, subtracting $J_0 - 4n$ from the whole buffer trajectory preserves nonnegativity—the buffer can decrease by at most $4n$ in n steps—and strictly decreases the root mean. Hence an optimum has a finite exact occupancy-LP representation

$$\min\{c^T u : Au = b, u \geq 0\},$$

with rational A, b, c .

Exact rational primal–dual certificates give the optimal values $\mathcal{B}_2, \mathcal{B}_3, \mathcal{B}_4$. Now fix the optimal face

$$Au = b, \quad u \geq 0, \quad c^T u = \mathcal{B}_n.$$

For a history h , let $q_h^T u = \mathbb{E}B_h$. Exact rational dual certificates for the secondary programs

$$\max q_h^T u$$

prove

$$\mathbb{E}B_{(4)} \leq \mathcal{B}_2 \quad (n = 3),$$

and

$$\mathbb{E}B_{(4)} \leq \mathcal{B}_3, \quad \mathbb{E}B_{(4,4)} \leq \mathcal{B}_2 \quad (n = 4).$$

But every corresponding subtree is feasible for its remaining horizon, so the reverse inequalities hold by definition of $\mathcal{B}_r$. Therefore all three inequalities are equalities, proving (3)–(4).

The certificates are checked using rational arithmetic only; no numerical optimizer enters the verification.

Finally insert $\Delta_h = \Delta_{h4} = 0$ into (2) with $s = 1$. The $x = 4$ term vanishes identically and the remaining terms give (5)–(6). $\square$

The realised line pays nothing extra. The bill belongs to the fan.